CardioPredict — Pandas & Data Exploration

Practical Coding Notes | YUVA Internship

1. Import Pandas

We use Pandas to work with tabular patient data.

Code

```
import pandas as pd
```

2. Read the Dataset

We created a practice dataset named Patients.csv and loaded it into a DataFrame.

The implementation is kept in sample.py.

Code

```
df = pd.read_csv("Patients.csv")
print(df)
```

3. Understand the DataFrame

A DataFrame is the table-like structure we work with in Pandas.

For our project, a row represents a patient record and a column represents a feature or field.

4. Explore the Dataset

These commands were practiced to understand the dataset before cleaning or Machine Learning.

Code

```
df.head()
df.tail()
df.shape
df.columns
df.info()
df.describe()
```

5. Select Columns and Rows

We practiced selecting one column, multiple columns, and a particular row.

Code

```
df["Age"]
df[["Age", "Name"]]
df.loc[0]
```

6. Filter Data

Filtering lets us ask questions about the dataset using conditions.

Code

```
df[df["Age"] > 50]
```

7. Basic Analysis

We practiced calculating an average, counting categorical values, checking missing values, and sorting.

Code

```
df["Age"].mean()
df["HeartDisease"].value_counts()
df.isnull().sum()
df.sort_values("Age")
```

8. What Each Command Helped Us Learn

head() → first rows.

tail() → last rows.

shape → number of rows and columns.

columns → available column names.

info() → structure, data types, and non-null information.

describe() → numerical summary.

loc → access rows.

filtering → select records that satisfy a condition.

mean() → calculate an average.

value_counts() → count values.

isnull().sum() → count missing values.

sort_values() → sort records.

9. Practice File

The examples from this learning stage are maintained in **sample.py**.

10. CardioPredict Connection

Patients.csv → Pandas DataFrame → explore the dataset → identify what needs cleaning → prepare the data for later Machine Learning.

The next major topic after this stage is Data Cleaning.